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MIL-PRF-85045/16D
5 November 2021
SUPERSEDING
MIL-PRF-85045/16C
17 June 2014

PERFORMANCE SPECIFICATION SHEET

CABLE, FIBER OPTIC, SINGLE (ONE) FIBER, CABLE CONFIGURATION TYPE 2 (OFCC), TIGHT BUFFER, CABLE CLASS SM AND MM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-85045.

CLASSIFICATION:

Fiber optic cable configuration type: 2 (optical fiber cable component [OFCC])

Fiber optic cable class: MM (graded-index, glass core and glass cladding, multimode)

SM (dispersion-unshifted, glass core and glass cladding, single-mode)

DESIGN AND CONSTRUCTION:

Fiber: Class MM fibers shall be in accordance with MIL-PRF-49291/6.

Class SM fibers shall be in accordance with MIL-PRF-49291/7.

Buffer diameter: 900±50 micrometers (µm)

Finished cable:

Dimensions and configuration: See figure 1.

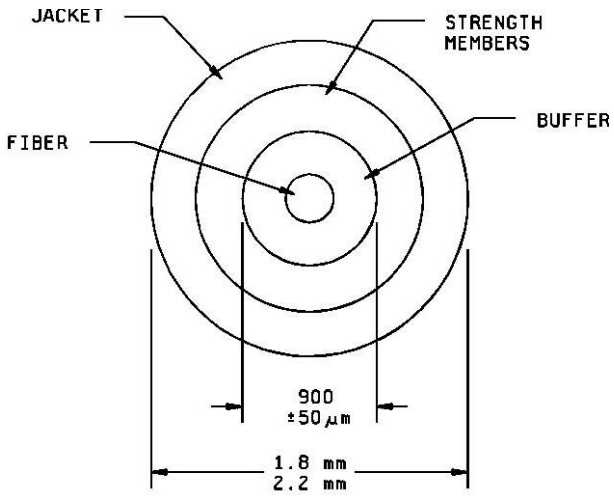


FIGURE 1. Tight buffered single fiber optical cable.

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Number of fibers: One

Concentricity: >0.8

Mass per unit length: ≤15 kilograms per kilometer (kg/km)

Short-term minimum bend diameter: Eight times the cable outer diameter. (The short-term minimum bend diameter shall be used in all environmental and mechanical tests that specify a cable minimum bend diameter.)

Long-term minimum bend diameter: Sixteen times the cable outer diameter.

Color: Slate for class MM fiber. Yellow for class SM fiber.

Jacket material: The jacket shall be composed of a low-halogen, low-smoke, low-toxicity polymer material.

Minimum continuous length: The minimum continuous length of all cables shall be not less than 0.5 km. If lengths less than 0.5 km are specified in the purchase order, a conformance inspection shall be performed on test specimens not less than 0.5 km in length from which the purchase order lengths are cut.

PERFORMANCE REQUIREMENTS: The individual performance requirements shall be as specified in MIL-PRF-85045 and this specification sheet, including the requirements identified in table I.

Optical properties:

Maximum attenuation rate: 4.5 decibels (dB)/km at 850±20 nanometers (nm), 2.0 dB/km at 1,300±20 nm for class MM fiber.

1.0 dB/km at 1,310±20 nm and 1,550±20 nm for class SM fiber.

Change in optical transmittance: Measurements shall be made at 1,300±20 nm.

Bandwidth measurements (multimode only): For this cable construction, the bandwidth is measured as part of the requirements of the optical fiber to be cabled. The bandwidth value does not change once the optical fiber is cabled. Since the bandwidth is provided as one of the cable measured parameters, the acceptable bandwidth for the optical fiber used in this cable (and by extension, the cable) is provided for informational purposes.

Bandwidth (multimode cables only): For cable with MM: 62.5/125 micron fiber: ≥200 megahertz (MHz)-km at 850±20 nm, ≥500 MHz-km at 1,300±20 nm for cable with MM fiber.

Mechanical properties:

Tensile loading and elongation: Applicable; tensile loading >270 newtons (N)

Dynamic bend: The tensile load shall be not less than 90 N.

Cyclic flexing: 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. Change in optical transmittance measurements shall be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Crush: The test load shall be 250 N at a rate of 250 N per minute.

Cable twist-bending: 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. The test load shall be 10 N. Change in optical transmittance measurements shall be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Cable jacket tear strength: Applicable, except the cable jacket tear strength shall be not less than 35 N per centimeter of jacket thickness for standard performance cable.

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Environmental properties:

Temperature range:

Operating: -28 to 85 °C

Nonoperating: -40 to 70 °C

Storage: -40 to 70 °C

Temperature cycling: Change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made over a period of 1 hour at the end of each temperature plateau.

Thermal shock: Applicable, except shall be performed at the operating temperature extremes.

Temperature humidity cycling: Change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made at the end of each temperature plateau.

Flaming smoke generation: D_m<225 when tested in the flaming condition in accordance with ASTM E662. The size and configuration of the test specimen shall conform to the 76- by 76-mm (3- by 3-inch) square specified in ASTM E662. The thickness of the test specimen shall be 2 mm (0.08 inch). The specimen shall be constructed by laying 76-mm (3-inch) lengths of OFCC adjacent to one another to form a 76-mm (3-inch) strip. Wire or cable clamps may be used to permanently hold the OFCC lengths in place. Longer OFCC lengths may be used if top or bottom clamps are used to hold the OFCCs. If top or bottom clamps are used, the length of OFCC exposed outside of the clamp(s) shall be 76 mm (3 inches).

Chemical properties:

Halogen content: <0.2 percent

Acid gas generation: Applicable. OFCCs shall be cut to approximately 2.5 centimeters long, and 16 OFCCs will be required to achieve a mass of 1.25 grams (minimum) to 2.5 grams (maximum) and shall be placed in a silica boat for test.

Identification Marking:

Part or identifying number (PIN):

M85045/16-01 (Multimode)

M85045/16-02 (Single-mode)

Qualification and conformance inspection: See table I. This table specifies applicable requirements, qualification inspections, and conformance inspections per this specification sheet. The Requirement Paragraph and Test Paragraph columns within this table reference MIL-PRF-85045. The Cable Length column of this table applies only to the qualification inspection. A reduced sample set is permitted for conformance inspection; refer to MIL-PRF-85045 for conformance inspection sample size.

Preconditioning: If preconditioning is required, it shall be performed as part of the cable fabrication process and cables shall be supplied already preconditioned. No preconditioning shall be performed as part of qualification

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TABLE I. Qualification and conformance inspection.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length [1/, 2/, 3/, 4/]	Conformance inspection
I	Visual and mechanical inspection	3.4, 3.9, 3.10	4.7.2	Three samples, 0.5 km each [5/]	A
	Attenuation rate	3.5.1	4.7.4.1	Three samples, 0.5 km each [6/]	A
III	Temperature cycling	3.7.1	[7/]	Two samples, 0.5 km each [6/] (one on reel, one off reel)	C
	Temperature humidity cycling	3.7.3	4.7.6.3	Two samples, 0.5 km each [8/]	C
	Storage temperature	3.7.4	4.7.6.4	Two samples, 0.5 km each [8/]	--
	Low temperature flexibility (cold bend)	3.6.3	4.7.5.3	Two specimens, 8 m each [9/]	B
	Cyclic flexing	3.6.4	4.7.5.4	Four specimens, 5 m each [10/] (Two specimens at each temperature)	--
	Crush	3.6.5	4.7.5.5	Two specimens, 5 m each [9/]	--
	Cable twist-bending	3.6.6	4.7.5.6	Four specimens, 5 m each [10/] (Two specimens for each temperature)	C
	Temperature life (life aging)	[7/]	[7/]	Two specimens, 300 m each [9/]	C
	Tensile loading and elongation	3.6.1	4.7.5.1	Two specimens, 150 m each [11/]	C
	Operating tensile loading	3.6.1.1	4.7.5.1.1	Two specimens, 150 m each [12/]	C
	Fungus resistance	3.8.4	4.8.4	Two specimens, 0.5 m each [9/]	--
	Cable element removability	3.6.18	4.7.5.18	Two specimens, 0.5 m each [9/]	C
	Flammability (60-degree angle)	3.7.12.1	4.7.6.12.1	Two specimens, 6 m each [9/]	C
IV	Thermal shock	3.7.2	4.7.6.2	One specimen, 0.49 km [6/] (on reel)	--
	Jacket self-adhesion or blocking	3.7.11	4.7.6.11	One specimen, 30 m [13/]	--
	Dynamic bend	3.6.2	4.7.5.2	One specimen, 150 m [14/]	B
	Corner bend	3.6.9	4.7.5.9	Two specimens, 5 m each [15/]	--
	OFCC kink	[7/]	[7/]	Two specimens, 1 m each [15/]	--

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TABLE I. Qualification and conformance inspection – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/ 2/ 3/ 4/	Conformance inspection
V	Cable jacket tear strength	3.6.14	4.7.5.14	Five flat extruded jacket material strips 16/	C
	Cable jacket material tensile strength and elongation	3.6.15	4.7.5.15	Five specimens 17/	--
	Flaming smoke generation	7/	7/	One specimen 18/	C
VI	Acid gas generation	3.8.1	4.8.1	One specimen, 1 m 19/	C
	Halogen content	3.8.2	4.8.2	One specimen, 1 m 19/	--
	Toxicity index	3.8.3	4.8.3	One specimen, 1 m 19/	C

NOTES:

1/ Tolerance on 0.5-km length is ±5 percent, provided that results are normalized to 1 km.
2/ Tolerance on shorter lengths is ±5 percent.
3/ A sample is the initial 0.5-km cable length.
4/ A specimen is a cable length that is cut from the initial 0.5-km cable length (i.e., the sample) or is cut from a longer previous cut length (i.e., cut from a longer specimen).
5/ The visual and mechanical inspection shall only be conducted on a 2-m section of each sample.
6/ The same samples as used in the visual and mechanical inspection shall be used.
7/ As stated under the applicable tests in this specification sheet.
8/ The same samples as used in the temperature cycling test shall be used.
9/ A specimen cut from each sample used in the temperature cycling test shall be used.
10/ Two specimens cut from each sample used in the temperature cycling test shall be used.
11/ A specimen cut from each specimen used in the temperature life test shall be used.
12/ The same specimen as used in the tensile loading and elongation test shall be used.
13/ A specimen cut from the specimen used in the thermal shock test shall be used.
14/ A specimen cut from the specimen used in the thermal shock test or one of the specimens used in the tensile loading and elongation test shall be used.
15/ Specimens cut from the specimen used in the thermal shock test shall be used.
16/ Flat extruded jacket material strips (i.e., strips of flat extruded material with the same composition as the cable jacket and of sufficient dimensions in which dumbbells can be cut) prepared as specified in 4.7.5.14 and obtained from flat extruded material that underwent the thermal shock test shall be used.
17/ Flat extruded jacket material as specified in 4.7.5.15 shall be used in this test.
18/ Use 76-mm (3-inch) lengths of cable adjacent to one another to form a 76-mm (3-inch) strip. For a 2-mm (0.08-inch) diameter cable, an approximate length of 3 m is needed.
19/ A specimen cut from one of the samples used in the temperature cycling test or specimen used in the thermal shock test shall be used.

Qualification and conformance by similarity: Manufacturers who produce products for MIL-PRF-85045/17 or MIL-PRF-85045/18 and this specification sheet, are qualified under MIL-PRF-85045/17 or MIL-PRF-85045/18, and whose product produced to this specification sheet passes the tests and inspections specified in table II, are qualified under this specification sheet. This qualification by similarity is applicable if the product produced to this specification was used in the production of the already qualified product under MIL-PRF-85045/17 or MIL-PRF-85045/18. Testing may be performed on a single length of cable with a minimum length of 0.5 km.

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TABLE II. Qualification and conformance by similarity.

Group	Inspection	PIN		Conformance inspection 3/
		M85045/16-01 1/	M85045/16-02 2/	
I	Size inspection	X	X	A
	Attenuation rate	X	X	A
III	Temperature cycling	X	X	C
	Low temperature flexibility (cold bend)	X	X	B
IV	Thermal shock	X	X	--
	OFCC kink	X	X	--
V	Flaming smoke generation	X	X	C

NOTES:
1/ Where manufacturers are qualified for M85045/17-01P or M85045/18-01P.
2/ Where manufacturers are qualified for M85045/17-02P or M85045/18-02P.
3/ All conformance inspections required for all qualified PINs.

Intended use: This cable is intended for use as jumper cordage and pigtails for fiber optic components. This cable is intended for use inside of protected enclosures and is not intended for installation in the overheads or cableways.

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
NASA – NA

Preparing activity:
Navy – SH
(Project 6015-2020-002)

Review activities:
Army – AR, AV, MI
Navy – EC, YD
Air Force – 02, 19, 70
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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